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Effect of an orally applied herbal immunomodulator on cytokine induction and antibody response in normal and immunosuppressed mice

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Summary

The influence of the oral administration of a herbal immunomodulator, consisting of an aqueous-ethanolic extract of the mixed herbal drugs Thujae summitates, Baptisiae tinctoriae radix, Echinaceae purpureae radix and Echinaceae pallidae radix, on cytokine induction and antibody response against sheep red blood cells was investigated in mice. The treatment of the animals with the extract caused no enhancement of the cytokine titers in the serum. Spleen cells isolated from the treated mice, however, produced higher amounts of IL-2, IFNγ and GM-CSF *ex vivo* in comparison to spleen cells isolated from control animals, especially after additional stimulation by lipopolysaccharides or concanavalin A. The application of the extract also triggered the production of IL-1 and TNFα by peritoneal macrophages *ex vivo*. The influence of the herbal extract on the antibody response was examined by the plaque forming cell assay. The administration of the extract caused a significant enhancement of the antibody response against sheep red blood cells, inducing an increase in the numbers of splenic plaque forming cells and the titers of specific antibodies in the sera of the treated animals. In mice, immunosuppressed by old age or additional treatment with hydrocortisone, the therapy with the extract resulted in a normalization of the antibody response against sheep red blood cells.

Key words: Herbal immunomodulator, *Echinacea* spp., *Baptisia* spp., *Thuja* spp., peroral application, PFC response in mice, cytokine induction

Introduction

Extracts from *Echinacea purpurea* (L.) MOENCH, *Echinacea pallida* (NUTT.) NUTT., *Baptisia tinctoria* L. or *Thuja occidentalis* L. are known for their immunomodulating activity. In human therapy they are applied especially in order to overcome weakness of the immune system and to prevent and to treat infections of the upper respiratory apparatus (Wagner and Wiesenauer, 1995). Their activity was not only proven in clinical studies but also in pharmacological test models where the immunomodulating activity of the extracts and their high molecular fractions was experimentally confirmed *in vitro* and *in vivo* (Bauer and Liersch, 1998; Harnischfeger, 1998; Staesche and Schleinitz, 1998; Wagner et al., 1985; Vömel, 1984; Vömel, 1985; Beuscher et al., 1995; Beuscher et al., 1990; Gohla et al., 1988). *In vivo*, test fractions were mostly applied *i.v.* or *i.p.* The relevance of these experimental data for human therapy, where low dosages of extracts are commonly applied orally in form of liquids or tablets, is being discussed controversially. Therefore we started experiments on the immunological effects of these extracts after peroral application (Bodinet, 1999; Bodinet and Freudenstein, 1999).

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One aim of this study was to clarify whether a perorally administered herbal immunomodulator exerts its action through systemic or local effects. Therefore we investigated its influence onto the cytokine production by quantifying cytokine concentrations in blood and by determining cytokine production of isolated immune cells *ex vivo*.

Since a temporarily weakened immune system is one main indication for the therapy with herbal immunomodulators (Wagner and Wiesenauer, 1995) we were also interested in investigating to what extent these herbals are able to restore decreased immune parameters in immunosuppressed animals. For that reason we tested the effects of a herbal extract on the antibody response against sheep red blood cells (SRBC) in mice immunosuppressed by old age or by treatment with hydrocortisone. The plaque forming cell (PFC) assay was used as test system, because it does not only consider one single immune parameter, but reflects all aspects of an adequate immune response (Holsapple et al., 1995).

Since it has been shown in animal experiments that combinations of plant extracts exert higher effects than extracts of single plants (Wagner and Jurcic, 1991), we used an extract of a mixture of immunomodulating plants for our investigations.

The aqueous-ethanolic extract of a mixture of Thujae summitates, Baptisiae tinctoriae radix, Echinaceae purpureae radix and Echinaceae pallidae radix was perorally applied to mice in dosages that are comparable to those in human therapy.

Materials and Methods

Plant material and extraction

The medicinal drugs were purchased from the following companies: Thujae occidentalis herba (G00561) from Alfred Galke GmbH (Germany), Baptisiae tinctoriae radix (G00345) from Paul Müggenburg GmbH & Co (Germany), Echinaceae purpureae radix (F01007) from Dieter Müller (Germany) and Echinaceae pallidae radix (G00205) from Berghof-Kräuter GmbH (Germany). The drugs were checked for identity and purity using the specifications of the Quality Control Department of Schaper & Brümmer.

Extraction was performed as described in an earlier publication (Bodinet and Freudenstein, 1999). In short: A mixture of Herba Thujae occidentalis, Baptisiae tinctoriae radix, Echinaceae purpureae radix and Echinaceae pallidae radix was exhaustively percolated with 30% EtOH according to the manufacturing specification for Esberitox® N (drug solvent relation 1:11). The extract was characterized by TLC and HPLC (Bodinet and Freudenstein, 1999). A part of the extract was concentrated under reduced pressure and freeze-dried.

HPLC

HPLC was carried out with Spherisorb ODS I 250 mm * 4 mm, particle size 5 µm; gradient system: mobile phase A: C_2H_3N with 1% 0,1 N H_3PO_4 and mobile phase B: H_2O with 1% 0.1 N H_3PO_4, gradient: 0–20.0 min: 5 $\rightarrow$ 25% A with total flow 1.0 ml/min, 20.0–20.1 min: 25 $\rightarrow$ 100% A with total flow 1.0 ml/min, 20.1–35.0 min: isocratic 100% A with total flow 1,5 ml/min, 35.0–35.1 min: 100 $\rightarrow$ 5% A with total flow 1.5 ml/min, 35.1–50.0 min: isocratic 5% A with total flow 1.5 ml/min; injection volume: 30 µl; UV detection at 254 nm and 330 nm. The corresponding HPLC profiles are shown in Fig. 1 and 2.

Animals

NMRI and C3H/HeJ-mice were obtained from Charles River (Germany) and housed in our veterinary section under standard conditions with food and water ad libitum.

Treatment

The lyophilized extract was administered to mice via the drinking water (autoclaved tap water) or with the diet. Based on the average food consumption a special diet was produced by Altromin International (Germany), containing the extract homogeneously admixed to the standard diet (Altromin 1314). The consumption of drinking water and food was measured daily by weighing the remains allowing the determination of the daily extract dose the animals had ingested on average.

Measurement of changes in cytokine production

Male NMRI-mice (4 weeks old, 20–30g) were distributed randomly on 2 groups of 5 animals each for each test day. The extract was applied perorally via the drinking water for 1–9 days. The mice in the control group received drinking water without additions. 24 hrs after the last application the animals were sacrificed.

Blood was collected by heart puncture. Serum was prepared according to standard methods and stored at –70 °C. The concentration of IL-1β, IL-2, IFNγ, TNFα and GM-CSF in the serum was quantified using commercial ELISAs (Genzyme Duo-Set mouse IL-1β, IL-2, IFNγ, TNFα; Endogen Elisa Set Mouse GM-CSF).

The spleens were isolated, washed with cell culture medium and single cell suspensions were prepared. After the last washing step the cells were suspended in RPMI containing 10% heat-inactivated fetal calf serum, 2-mercaptoethanol (50 µmol/ml), penicillin (100 U/ml) and streptomycin (100 µg/ml). The cell suspension was adjusted to a cell number of 5×10^6 cells/ml. Cell suspensions of each group were pooled and plated in 24-well microtiter plates (1 ml/well). Another 1 ml of cell culture medium or a solution of LPS

(E. coli 0128: B12, 10 µg/ml) or ConA (Lectin of *Canavalia ensiformis* 2 µg/ml) were added to each well. Plates were incubated at 37 °C and 5% CO_2. After 48 hrs supernatants of 4 parallel wells were collected, pooled, centrifuged and frozen at –70 °C. The amount of IL-2, IFNγ and GM-CSF in the supernatant was quantified using commercial ELISAs (see above).

Resident peritoneal cells were harvested by peritoneal lavage. The peritoneal liquid of the 5 mice from each group was pooled and centrifuged for 10 min at 1000 rpm and 4 °C. After washing the cells were suspended in RPMI containing 10% heat-inactivated fetal calf serum, 2-mercaptoethanol (50 µmol/l), penicillin (100 U/ml) and streptomycin (100 µg/ml). The cell suspension was adjusted to a density of 1×10^6 cells/ml. Smears of the cell suspensions were stained with Hemacolor Quick Dye (Merck), counted and differentiated microscopically. The cell suspensions consisted to 50–70% of macrophages. 1 ml/well of the cell suspensions were seeded in 24-well microtiter plates. After 4 hrs incubation at 37 °C and 5% CO_2 the non-adherent cells were removed by washing twice with cold cell culture medium. 2 ml of cell culture medium or 2 ml of a solution of LPS (10 µg/ml) were added to 6 wells each.

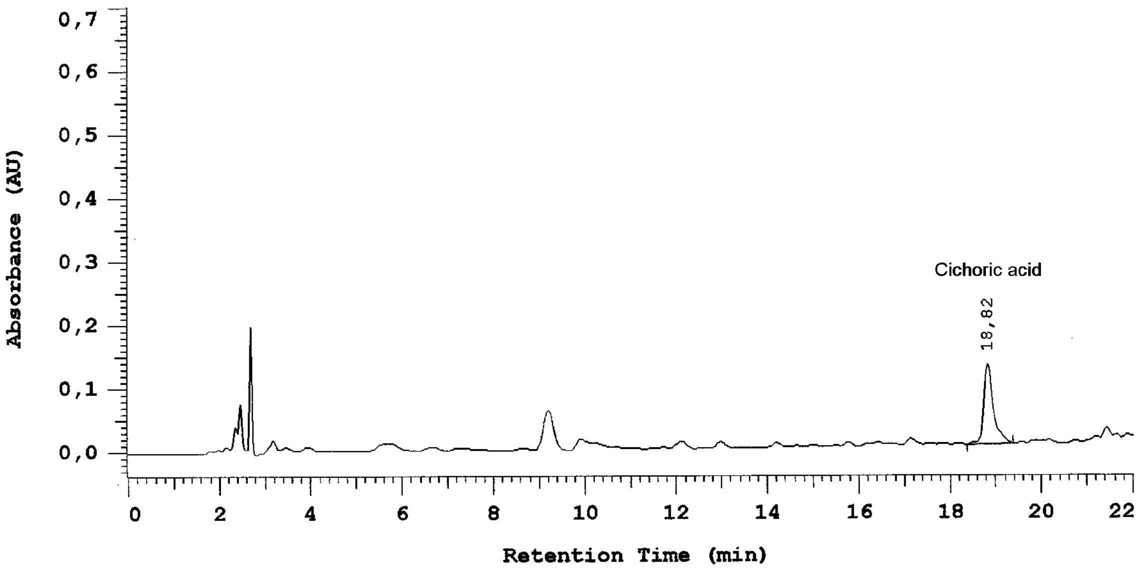

Fig. 1. HPLC profile of the herbal extract, UV absorbance at 254 nm (HPLC conditions see Materials and Methods).

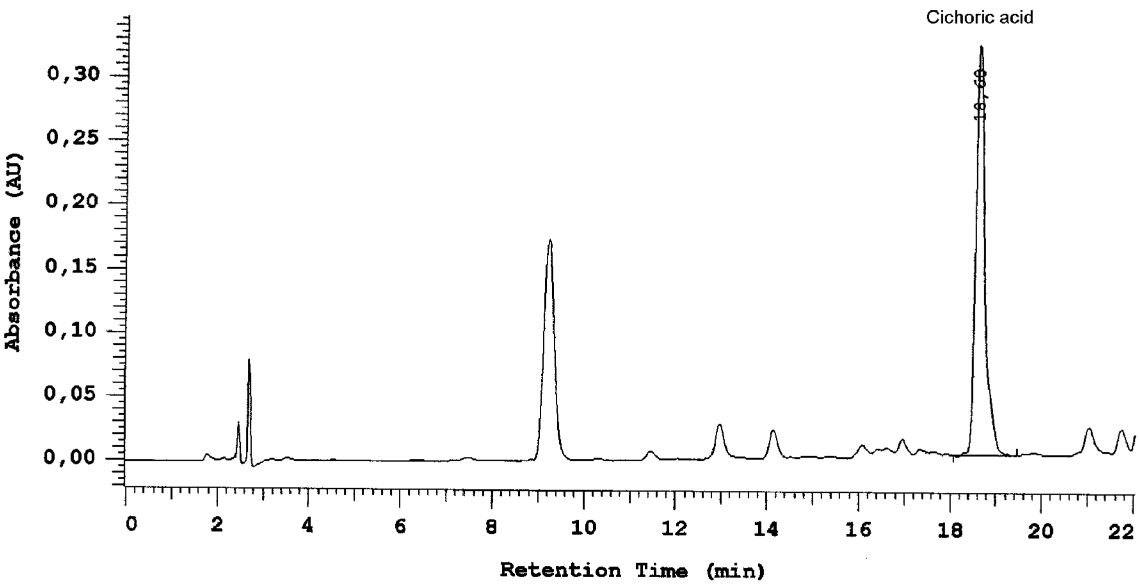

Fig. 2. HPLC profile of the herbal extract, UV absorbance at 330 nm (HPLC conditions see Materials and Methods).

The plates were incubated at 37 °C and 5% CO_2. After 24 hrs the supernatants from 6 parallel wells were removed, pooled, centrifuged and frozen at –70 °C. The concentrations of IL-1β and TNFα in the supernatant were quantified using commercial ELISAs (see above).

Antibody response to sheep red blood cells

Tests were performed with 12 month old C3H/HeJ-mice (25–30g) or young male NMRI-mice (4 weeks old, 20–30g), immunosuppressed by additional treatment with hydrocortisone.

C3H/HeJ-mice were distributed randomly on 2 groups of 10 animals. The extract was applied perorally via the drinking water. The mice in the control group received drinking water without additions.

Each 6 NMRI-mice per group received the standard diet (group 1), the standard diet plus hydrocortisone (group 2), or the special diet, containing the lyophilized extract, and hydrocortisone (group 3). Hydrocortisone (25 mg/kg body weight) was applied i.p. 2 days before immunization.

The extract was applied for 7 consecutive days. On the fourth day the mice were immunized by i.p. injection of 0.2 ml of a 3% suspension of SRBC in physiological NaCl. 4 days later the number of antibody producing cells per spleen was determined with the plaque forming cell assay as described earlier (Bodinet and Freudenstein, 1999).

Additionally the spleen weights, number of cells per spleen, hemagglutination titers and blood cell counts were determined for each animal individually (for details see Bodinet and Freudenstein, 1999).

Statistics

The results were expressed as mean ± standard deviation (S.D.). Comparison between the values of the treated animals and the controls were performed using the Student's *t*-test with a limit of significance at $p = 0.05$.

Results

There were no recognizable differences in the cytokine titers (IL-2, IFNγ, GM-CSF, IL-1 and TNFα) between the sera of the NMRI-mice that had been treated perorally with the herbal extract for a period of 1–9 days and the sera of the control animals. The concentrations of the cytokines in the sera of both groups ranged around the detection limit of the respective assays.

The spleen cells of the NMRI-mice, however, that had been treated with the extract for 1–9 days produced higher amounts of IL-2, GM-CSF and IFNγ *ex vivo* in comparison to those from the control animals. The cytokine production could be further enhanced by an additional stimulation with LPS or ConA, whereby the spleen cells from the treated animals reacted to the co-stimulation considerably more strongly and produced by far higher amounts of IL-2, GM-CSF and IFNγ. Differences in the cytokine production between the spleen cells from the verum and from the control animals were already recognizable after the first treatment day and remained so until day 9, however, with a tendency towards attenuation. The clearest differences in cytokine production between spleen cells of verum and control animals appeared after 3-day administration of the extract. The spleen cells from the treated animals produced approximately 7 times more IL-2, 13 times more GM-CSF and 60 times more IFNγ than the spleen cells from the control animals (Table 1).

The peritoneal macrophages of the NMRI-mice, which had been treated with the herbal extract, produced more IL-1 and TNFα than the peritoneal macrophages from the control mice, particularly after co-stimulation with LPS *ex vivo*. The differences were recognizable from the 1st day of treatment regarding IL-1 and from the 3rd day regarding TNFα production (Table 2). The effects remained unchanged during the following treatment period.

In the test series with 12 month "old" C3H/HeJ-mice the extract was continuously applied via the drinking water for 7 days. The mean daily dose was calculated by the drinking water consumption with 309 µl extract/kg body weight daily. The treatment led to a sig-

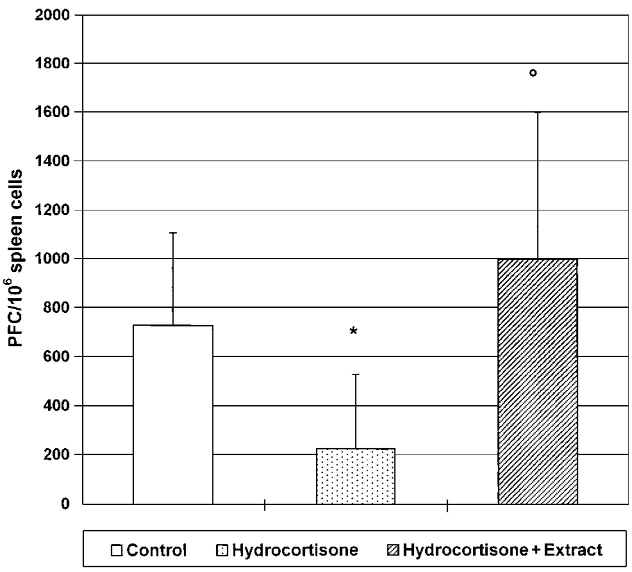

Fig. 3. Influence of the peroral application of the herbal extract on the PFC response to SRBC in immunosuppressed NMRI-mice.
The extract was applied via the food at an average dose of 230 µl extract/kg body weight daily on 7 consecutive days. Mice were immunized with SRBC on the 4th day. Hydrocortisone (25 mg/kg) was applied i.p. on 2 days before immunization. PFC-assay was performed on day 8.
n = 6 mice/group *$p < 0.05$ vs. controls (group 1) ° $p < 0.05$ vs. HC group (group 2) (Student's t-test).

Table 1. Influence of the herbal extract on the IL-2-, INFγ- and GM-CSF- production of mouse spleen cells *ex vivo*.

	Interleukin-2 (pg/ml)		Interferon γ (pg/ml)		GM-CSF (pg/ml)	
groups	control	extract	control	extract	control	extract
cell culture medium	42*	68*	56*	3430 ± 1078	6.9*	88.5 ± 30.4
+ Con A	1371 ± 86	2500 ± 141	19130 ± 3578	85418 ± 27319	374 ± 56	660 ± 38.2
+ LPS	192 ± 168	454 ± 164	136700*	184470*	656.5 ± 74.2	1523 ± 382.5

The lyophilized herbal extract was applied via the drinking water at an average dose of 260 µl extract/kg body weight daily on 3 consecutive days. The spleen cells were harvested 24 hrs after the last administration and incubated with or without ConA or LPS for 48 hrs at 37 °C in a 5% CO_2 atmosphere. The cytokine titers in the supernatants were quantified by ELISA.
* values out of standard range

Table 2. Influence of the herbal extract on the IL-1β and TNFα production of mouse peritoneal macrophages *ex vivo*.

	Interleukin-1 (pg/ml)		Tumor Necrosis Factor α (pg/ml)	
groups	control	extract	control	extract
cell culture medium	30.3 ± 9	56.8 ± 18.5	253 ± 46	426 ± 26
+ LPS	25.5 ± 11	272 ± 123	1130 ± 284	4493 ± 1232

The lyophilized herbal extract was applied via the drinking water at an average dose of 260 µl extract/kg body weight/day on 3 consecutive days. The peritoneal macrophages were prepared 24 hrs after the last administration and incubated with or without LPS for 24 hrs at 37 °C in 5% CO_2 atmosphere. The cytokine titers in the supernatants were quantified by ELISA.

Table 3. Influence of the herbal extract on the antibody response against SRBC in "old" C3H/HeJ mice.

Group	PFC/10^6 spleen cells	Hemagglutination titer	Spleen weight (g)	Cell yield/spleen
Control	917 ± 140	1189 ± 242	0.14 ± 0.02	7.4 ± 2 × 10^7
Extract	1308 ± 475 *	1920 ± 1491	0.21 ± 0.08 *	11.6 ± 3.1 × 10^7 *

The lyophilized herbal extract was applied to 12 month old C3H/HeJ-mice via the drinking water at an average dose of 309 µl extract/kg body weight daily on 7 consecutive days. Mice were immunized with SRBC on the 4th day. PFC-assay was performed on day 8.
n = 10 mice/group *$p < 0.05$ vs. controls (Student's t-test).

Table 4. Influence of the herbal extract on the antibody response against SRBC in immunosuppressed NMRI-mice.

Groups	Spleen weight (g)	Cell yield/spleen	PFC/10^6spleen cells	Hemagglutination titer
1. Control	0.19 ± 0.02	6.1 ± 0.1 × 10^7	730 ± 379	1283 ± 1140
2. Hydrocortisone	0.15 ± 0.02 *	3.8 ± 1.7 × 10^7 *	231 ± 303 *	378 ± 508
3. Hydrocortisone + Extract	0.17 ± 0.01 °	5.9 ± 0.8 × 10^7 °	1004 ± 606 °	1760 ± 944

The herbal extract was applied via the food at an average dose of 230 µl extract/kg body weight daily on 7 consecutive days. Mice were immunized with SRBC on the 4th day. Hydrocortisone (25 mg/kg) was applied i.p. on 2 days before immunization. PFC-assay was performed on day 8.
n = 6 mice/group *$p < 0.05$ vs. controls (group 1) ° $p < 0.05$ vs. HC group (group 2) (Student's t-test).

nificant increase in the spleen weight, the total cell number and the number of antibody-producing cells in the spleen. In addition a slight shift to the left of the blood cell counts was found in the animals of the verum group (data not shown). The hemagglutination titers in the serum of these animals were also elevated, but showed more variability (Table 3).

The treatment of NMRI-mice with hydrocortisone led to a significant immunosuppression expressed in a decrease in the spleen weight, the total number of cells per spleen and the extent of the antibody response against SRBC (number of PFC/10^6 spleen cells and hemagglutination titers). The effects induced by hydrocortisone were compensated by the treatment with the extract supplied via the special diet. Spleen weights, number of cells per spleen, number of plaque forming cells and anti-SRBC-serum titers of the treated animals (group 3) were significantly enhanced in comparison to the animals that received only the standard diet (group 2) and reached standard levels again (group 1). A summary of the results is shown in table 4 and figure 3. The blood cell counts did not show any changes on the 5^{th} day after hydrocortisone application (data not shown).

Discussion

In our experiments the oral application of the herbal extract in healthy NMRI-mice did not induce an increase in the concentration of cytokines in the blood. The isolated spleen cells of the treated mice, however, *ex vivo* produced higher amounts of IL-2, GM-CSF and IFNγ and their peritoneal macrophages produced more IL-1 and TNFα respectively, compared to the cells isolated from the control animals. The differences were seen in particular, if there was an additional *ex vivo* stimulation of the cells with LPS or ConA, mimicking the action of microbial antigens, such as bacterial lipopolysaccharides, in the case of infections.

One may deduce from these results that the peroral administration in clinically relevant doses does not induce any systemic increase in the cytokine titers, but could probably cause a local activation of the cytokine-producing cells in the sense of "Priming". That means that after local contact with the immunmodulatory active components of the extract in the gut or in the upper part of the bronchial system, cells are enabled to react to an additional stimulation with an increased cytokine secretion (Brough-Holub and Kraal, 1997; Brough-Holub et al. 1995). Under physiological aspects such a local modulatory cytokine induction is to be assessed more positively than a systemic one, which in case of an overexpression could cause considerable side effects (Azemar, 1997; Dingermann and Zündorf, 1998; Roeb and Rose-John, 1996).

In order to simulate the state of a weakened immune system in our test model we used "old" mice or young mice treated with hydrocortisone. It is well known that the capacity of the immune system decreases with age. Yoshii et al. (1989) and Shibata et al. (1990) e.g. showed that in "old" mice (16 months) the antibody response against SRBC was reduced by 80% in comparison with young animals. As a cause for this, a T-cell deficiency was presumed. In our experiments the smaller spleen weights and the reduced total numbers of cells in the spleen of the "old" in comparison to young mice indicated an immunosuppressed state.

The immunosuppression induced by hydrocortisone is also well investigated (Gillis et al., 1979; Haussmann, 1998; Liang et al. 1997). In our model the effect of hydrocortisone could be shown by the reduction of the spleen weights, the total numbers of cells in the spleen and the intensity of the antibody response against SRBC.

Through peroral treatment with the extract the antibody response against SRBC could be increased significantly in old C3H/HeJ-mice. In comparison to the controls the verum animals had a significantly higher spleen weight and a significantly increased number of cells in the spleen. The therapy with the herbal immunomodulator probably caused a compensation of the T-cell deficiency, which could lead to the re-increase in the spleen weight and the total cell number in the spleen and finally to the increase in the antibody response against SRBC. Yoshii et al.(1989) and Shibata et al. (1990) experimentally confirmed that the T-cell deficiency in "old " mice was induced by a reduction of the IL-2 production and in part also by a reduction of the IL-1-production. FRASCA et al. (1985) verified that the function of the helper T-cells in old mice could be improved by an IFNγ-induced increase in the IL-2 production. In further experimental investigations YOSHII et al. (1989) showed that it was possible to induce a restoration of the decreased T-cell-dependent antibody response by restoration of the IL-2 and IL-1-production.

Interestingly the therapy with the extract did not only enhance the numbers of plaque forming cells and hemagglutination titers, but also had an influence on the spleen weight and the number of cells in the spleen. These parameters were not influenced in healthy young mice (Bodinet and Freudenstein, 1999). Obviously the extract does not influence these immune parameters when they are in their "normal", balanced state but only intervenes in case of a misbalance.

The immune response, suppressed through treatment with hydrocortisone, was also normalized after administration of the extract. Since the hydrocortisone-induced immunosuppression is also associated with a decrease in the number of T-lymphocytes and a decrease in the IL-2 production, this result implies that through peroral therapy with the herbal immunomodulator a relevant substitution had occurred.

Our results are particularly important under therapeutic aspects, since the weakened immune system ranks high among the main indications for herbal immunostimulants and it was possible to show in two different models (old or hydrocortisone-treated mice), that a weakened immune system can regain its normal capacity through therapy with the investigated immunomodulator.

The immunological potential of the plants we used for extraction, is well known. It could be shown that a polysaccharide fraction from Thujae occidentalis herba (molecular mass > 300000 Dalton) induced an activation of the phagocytosis of polymorphnuclear granulocytes and had also a mitogenic activity on leukocytes of the peripheral human blood (Gohla et al. 1988). Offergeld et al. (1992) showed that the Thuja-polysaccharides also stimulated the cytokine production of human peripheral blood lymphocytes. Moreover, podophyllotoxin, another constituent of Thuja occidentalis, stimulated in extremely low dosages the IL-1 and IL-2 production by human monocytes and lymphocytes (Harnischfeger, 1998; Zheng et al., 1987).

The immunomodulating effects of Baptisiae tinctoriae radix are associated with polysaccharides and glycoproteins (Staesche and Schleinitz, 1998). Wagner et al. (1985) isolated polysaccharide fractions with molecular masses of 25000–500000 Dalton. In mice, these fractions caused a raised Carbon-Clearance-rate and in human granulocytes an increase in the phagocytosis-index. In addition to polysaccharides, Beuscher et al. (1989) isolated and characterized glycoproteins which showed a mitogenic activity in the proliferation assay with spleen cells, induced an intensification of the antibody production and increased the IL-1 production of mouse macrophages. In an infection model with *Pseudomonas aeruginosa* it could be shown that the i.p. application of the polymer compounds from *Baptisia tinctoria* caused a significant increase in the survival rate of the treated mice (Beuscher et al. 1990).

Caffeic acid derivatives, alkamides, polysaccharides and also glycoproteins are discussed as the immunologically active principles of *Echinacea*. Their activities have been proven in numerous investigations *in vitro* and *in vivo* (Bauer and Liersch, 1998; Blaschek et al., 1998).

It has been shown in animal experiments that combinations of plant extracts exert higher effects than extracts of single plants (Wagner and Jurcic, 1991). It may be assumed that these synergism-effects play a role for the overall immunological activity of the herbal immunomodulator we investigated. This has to be cleared in further studies.

References

Azemar M (1997) Immunstimulantien: Sind sie hilfreich oder gefährlich? Pharm. Ztg. 142: 11–21

Bauer R, Liersch R (1998) Echinacea. In: Hänsel R, Keller K, Rimpler H, Schneider,G editors. Hagers Handbuch der Pharmazeutischen Praxis, 5. neubearbeitete Auflage, Bd. 5. Springer: Berlin, Heidelberg, New York, 1–34

Beuscher N, Scheit K-H, Bodinet C, Kopanski L (1998) Immunologically active glycoproteins of Baptisia tinctoria. Planta Med: 55: 358–63

Beuscher N, Scheit K-H, Bodinet C, Egert D (1990) Modulation of host resistance by polymeric substances from Baptisia tinctoria and Echinacea purpurea. In: Mashihi, K.N., Lange, W., editors. Immunotherapeutic Prospects of Infectious Diseases. Springer: Berlin-Heidelberg: 59–63

Beuscher N, Bodinet C, Willigmann I, Egert D (1995) Immunmodulierende Eigenschaften von Wurzelextrakten verschiedener Echinacea-Arten. Z. Phytother. 16: 157–66

Blaschek W, Döll M, Franz G (1998) Echinacea-Polysaccharide. Z. Phytother. 19: 255–62

Bodinet C (1999) Immunpharmakologische Untersuchungen an einem pflanzlichen Immunmodulator. Dissertation, Greifswald, EMAU Greifswald

Bodinet C, Freudenstein J (1999) Effects of an orally applied aqueous-ethanolic extract of a mixture of Thuja occidentalis herba, Baptisiae tinctoriae radix, Echinaceae purpureae radix and Echinaceae pallidae radix an antibody response against sheep red blood cells in mice. Planta Med. 65: 695–9

Broug-Holub E, Persoons JHA, Schornagel K, Kraal G (1995) Changes in cytokine and nitric oxide secretion by rat alveolar macrophages after oral administration of bacterial extracts. Clin. Exp. Immunol. 101: 302–7

Broug-Holub E, Kraal G (1997) In vivo study on the immunomodulating effects of OM-85 BV on survival, inflammatory cell recruitment and bacterial clearance in Klebsiella pneumonia. Int. J. Immunopharmacol. 19: 559–64

Dingermann T, Zündorf I (1998) Das Immunsystem: Ziel und Quelle von Arzneistoffen. PZ Prisma 5: 28–34, 108–13

Frasca D, Adorini L, Landolfo S, Doria G (1985) Enhancing effect of IFN-gamma on helper T cell activity and IL 2 production. J. Immunol. 134: 3907–11

Gillis S, Crabtree GR, Smith KA (1979) Glucocorticoid-induced inhibition of T cell growth factor production. I. The effect on mitogen-induced lymphocyte proliferation. J. Immunol. 123: 1624–-31

Gohla SH, Haubeck HD, Neth RD (1988) Mitogenic activity of high molecular polysaccharide fractions isolated from the cupressaceae Thuja occidentale L. I. Macrophage-dependent induction of CD-4-positive T-helper (TH+) lymphocytes. Leukemia 2: 528–33

Harnischfeger G (1998) Thuja. In: Hänsel R, Keller K, Rimpler H, Schneider G, editors. Hagers Handbuch der pharmazeutischen Praxis, 5. neubearbeitete Auflage, Bd. 6. Springer: Berlin, Heidelberg, New York, 955–66

Haussmann W (1998) Wirkprinzip von Immunsuppressiva und Immuninduktoren. PZ Prisma 5: 175–86

Holsapple MP (1995) The Plaque-Forming Cell (PFC) Response in Immunotoxicology. In: Burleson GR, Dean JH, Munson AE, editors. Methods in Immuntoxicology, Vol. 1. Wiley-Liss: New York, 71–108

Liang HR, Vuorela P, Vuorela H, Hiltunen R (1997) Isolation and immunomodulatory effect of flavonol glycosides from Epimedium hunanense. Planta Med. 63: 316–9

Offergeld R, Reinecker C, Gumz E, Schrum S, Treiber R, Neth RH, Gohla SH (1992) Mitogenic activity of high molecular polysaccharide fractions isolated from the cuppressaceae Thuja occidentalis L. enhanced cytokine-production by thujapolysaccharid, g-fraction (TPSg). Leukemia 6: 189–91

Roeb E, Rose-John S (1996) Cytokines: biology and therapeutic relevance. Dtsch. Med. Wochenschr. 121: 803–9

Shibata K, Kobayashi T, Takayanagi N, Fujiwara M, Yamamoto I (1990) Immunomodulatory effect of a newly synthesized compound, TOK-8801 (N-(2-phenylethyl)-3,6,6-trimethyl-5,6-dihydroimidazo [2,1-b]thiazole-2-carboxamide) on antibody production in vivo and delayed-type hypersensitivity in mice. Int. J. Immunpharmacol. 12: 497–502

Staesche K, Schleinitz H (1998) Baptisia. In: Hänsel R, Keller K, Rimpler H, Schneider G, editors. Hagers Handbuch der Pharmazeutischen Praxis, 5. neubearbeitete Auflage, Bd. 4. Springer: Berlin, Heidelberg, New York, 461–7

Vömel T (1984) Einfluss eines unspezifischen Immunstimulans auf die Phagozytose von Erythrozyten und Tusche durch das retikulohistiozytäre System unterschiedlich alter, isoliert perfundierter Rattenlebern. Arzneim.-Forsch. 34: 691–5

Vömel T (1985) Der Einfluss eines pflanzlichen Immunstimulans auf die Phagozytose von Erythrozyten durch das retikulohistiozytäre System der isoliert perfundierten Rattenleber. Arzneim.-Forsch. 35: 1437–9

Wagner H, Jurcic K (1991) Immunologische Untersuchungen von pflanzlichen Kombinationspräparaten. Arzneim.-Forsch. 41: 1072–6

Wagner H, Wiesenauer M (1995) Phytotherapie. Gustav Fischer: Stuttgart, Jena, New York

Wagner H, Proksch A, Riess-Maurer I, Vollmar A, Odenthal S, Stuppner H, Jurcic K, Le Turdu M, Fang JN (1985) Immunstimulierend wirkende Polysaccharide (Heteroglykane) aus höheren Pflanzen. Arzneim.-Forsch. 35: 1069–75

Yoshii H, Tatsumi S, Matsuoka T, Suehiro S, Konishi JE, Yanagihara Y, Negoro S, Kishimoto S (1989) Immunopharmacological actions of an extract isolated from inflamed skin of rabbits inoculated with vaccinia virus (Neurotropin) – II. Restorative effect on immune responses through the recovery of IL-2 production in aging mice. Int. J. Immunpharmacol. 11: 534–50

Zheng QY, Wiranowska M, Sadlik JR, Hadden JW (1987) Purified podophyllotoxin (CPH-86) inhibits lymphocyte proliferation but augments macrophage proliferation. Int. J. Immunpharmacol. 9: 539–49

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